

4a. Option 1 - Create semantic model connections with Power BI desktop

Training.**tips**

Use this option if you have Power BI desktop installed. We cannot yet create DirectQuery mode models in the browser, so we'll create our connections in desktop and then publish them for further editing in the browser.

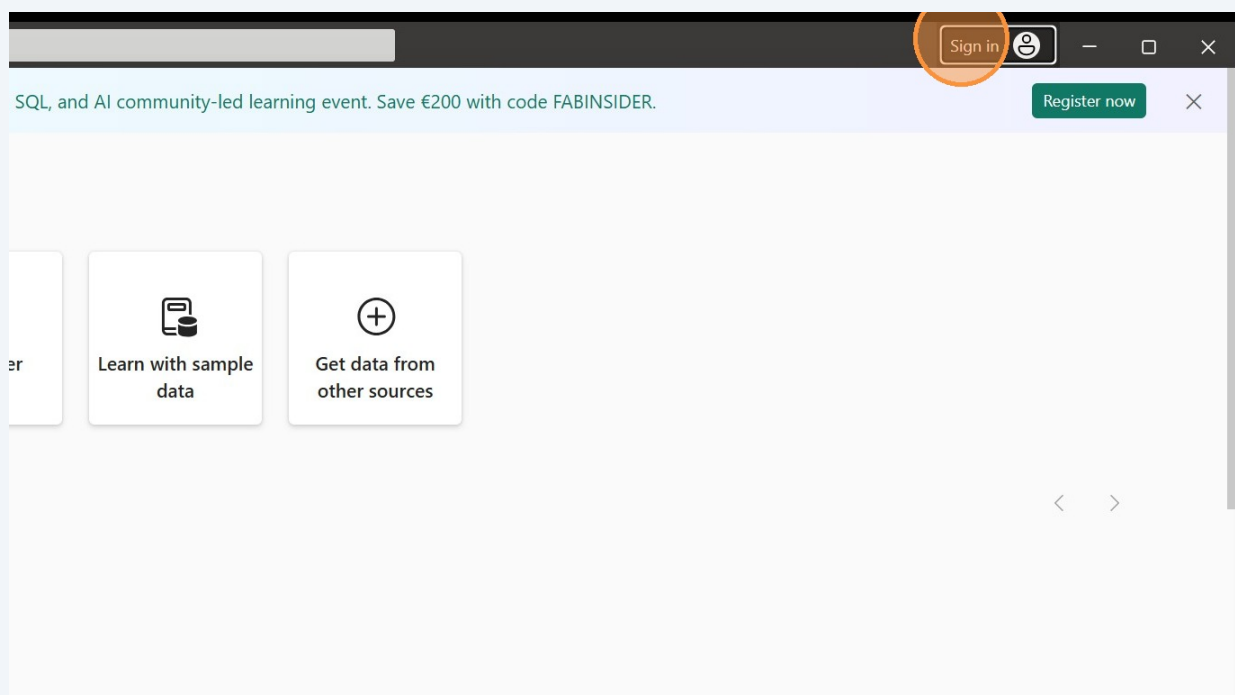
Sign in to Power BI Desktop



To connect to our workshop workspace, we need to be logged into Power BI desktop with our temporary workshop accounts, not our regular work accounts.

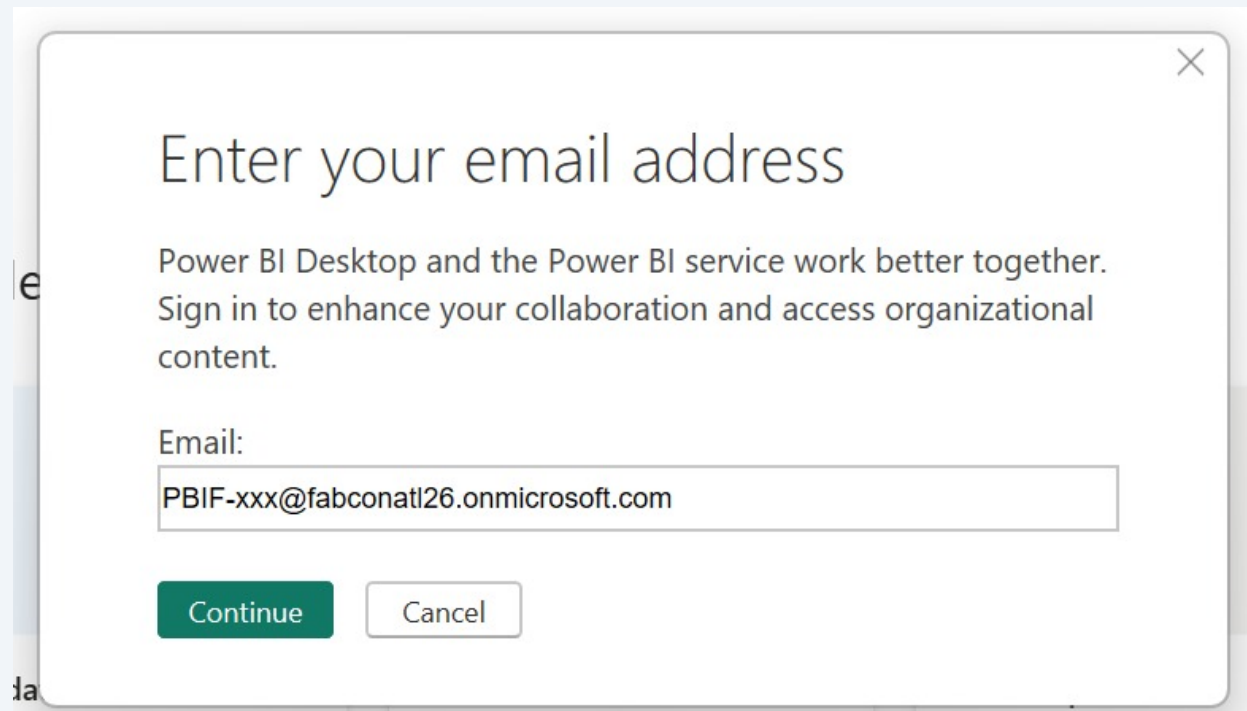
1

Open Power BI desktop and click "Sign in". If you're already signed in with another account, click on your profile image to show the dropdown and select "Sign in with a different account."



2

Enter your workshop email address and click "Continue". You'll have to enter your email address one more time and then the password.



The screenshot shows a dialog box titled "Enter your email address" with a close button (X) in the top right corner. The text inside the dialog box reads: "Power BI Desktop and the Power BI service work better together. Sign in to enhance your collaboration and access organizational content." Below this text is a label "Email:" followed by a text input field containing the email address "PBIF-xxx@fabconatl26.onmicrosoft.com". At the bottom of the dialog box are two buttons: "Continue" (a green button) and "Cancel" (a white button with a grey border).

Connect to your lakehouse - Import mode

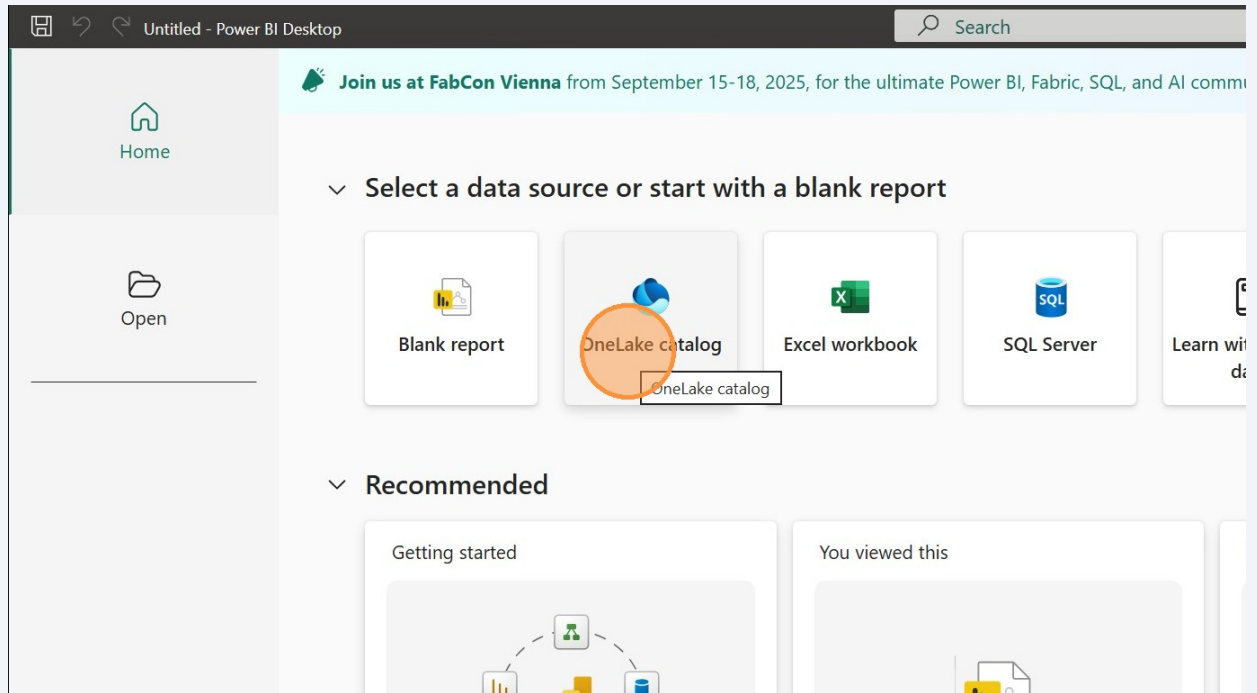


Tip! if you didn't have time to complete the dataflows lab to land your transformed data into your lakehouse, you can import from a shared lakehouse. Skip this section and move to the next section and import all **six** tables from the shared lakehouse.

3 We'll start by getting data from the lakehouse.

Click the "OneLake catalog" button.

If you're already in the Power BI canvas, there is a OneLake catalog button in the Home tab of the ribbon. Both buttons take us to the same place.



4

Navigate to your **lakehouse** in your workspace and click the name of the **lakehouse**. This window doesn't have a column to tell you what type of object each item is, but if you hover over the icon it will tell you.

	Name	Owner	Refreshed	Location	Endorsement
	fabcon info	Stephanie Bruno	—	Public	—
	FabCon Sessions	Stephanie Bruno	9/8/25, 12:03:23 PM	Public	—
	udf_fabcon_session_attendees	Stephanie Bruno	—	Public	—
	fabcon sessions dl2	Stephanie Bruno	9/10/25, 12:23:36 P...	Public	—
	fabcon_info	Stephanie Bruno	—	Public	—
	Fabric Chargeback Reporting	Stephanie Bruno	9/13/25, 12:07:50 ...	Fabric Chargeback R...	—
	Fabric Capacity Metrics	Stephanie Bruno	9/13/25, 12:15:04 ...	Microsoft Fabric Cap...	—
	skiina	Stephanie Bruno	—	Public	—

[Learn more about c](#)

Alert! On the next step, do not click the "Connect" button, rather click the down arrow next to the "Connect" button.

5

Don't just click the big green Connect button, because then you'll connect in Direct Lake mode. We want to import this data today. Click the down arrow next to the Connect button and then select "Connect to SQL endpoint."

	Owner	Refreshed	Location	Endorsement	Sensitivity
	Stephanie Bruno	—	Public	—	—
s	Stephanie Bruno	9/8/25, 12:03:23 PM	Public	—	—
ion_attendees	Stephanie Bruno	—	Public	—	—
dl2	Stephanie Bruno	9/10/25, 12:23:36 P...	Public	—	—
	Stephanie Bruno	—	Public	—	—
ck Reporting	Stephanie Bruno	9/13/25, 12:07:50 ...	Fabric Chargeback R...	—	—
Metrics	Stephanie Bruno	9/13/25, 12:15:04 ...	Microsoft Fabric Cap...		
	Stephanie Bruno	—	Public		

Connect to OneLake ⓘ

Connect to SQL endpoint ⓘ

Connect

Cancel

[Learn more about connection options](#) ↗



Tip! After clicking Connect, you may get an additional pop-up login that will take you to the browser. If so, go through the additional authentication.

6 Check the boxes next to the four tables you created:

- session speakers
- session tracks
- sessions
- speakers

The 'Select Related Tables' dialog is shown with the following tables selected:

- ☒ session speakers
- ☒ session tracks
- ☒ sessions
- ☒ speakers

The table preview on the right shows the following data:

ID	Name	Count
1001	Just Blindbaek	269
1001	Asgeir Gunnarsson	497
1005	Sander van de Velde	422
1007	Stephanie Bruno	74
1016	Marco Casalaina	308
1025	Brian Bønk	148
1027	Wolfgang Strasser	85
1030	Julie Koesmarno	268
1030	Robert Bruckner	405
1043	Mike Carlo	52
1058	Reitse Eskens	400
1066	Cristian Angyal	16
1068	Shabnam Watson	68
1071	Robert Bruckner	405
1071	Gwenaëll Bégo	221
1076	Monica Morehouse	1
1079	Erwin de Kreuk	26
1086	Kim Manis	43
1086	Wangui McKelvey	82
1088	John Morehouse	37
1089	Asgeir Gunnarsson	497
1091	Andy Cutler	116
1092	Kev Chant	41
1097	Elena Drakulevska	20

The 'Load' button is highlighted with an orange circle.

- 7 Leave this checked as "Import" and click the "OK" button.

Connection settings

You can choose how to connect to this data source. Import allows you to bring a copy of the data into Power BI. DirectQuery will connect live to this data source.

- ☒ Import
☐ DirectQuery

[Learn more about DirectQuery](#)



- 8 Notice the data being imported. Wait for this to finish.

Loading data...

-  gold date
Loading data...
-  gold session speakers
Loading data...
-  gold sessions
Loading data...
-  gold speakers
Loading data...



Connect to the shared lakehouse - Import mode

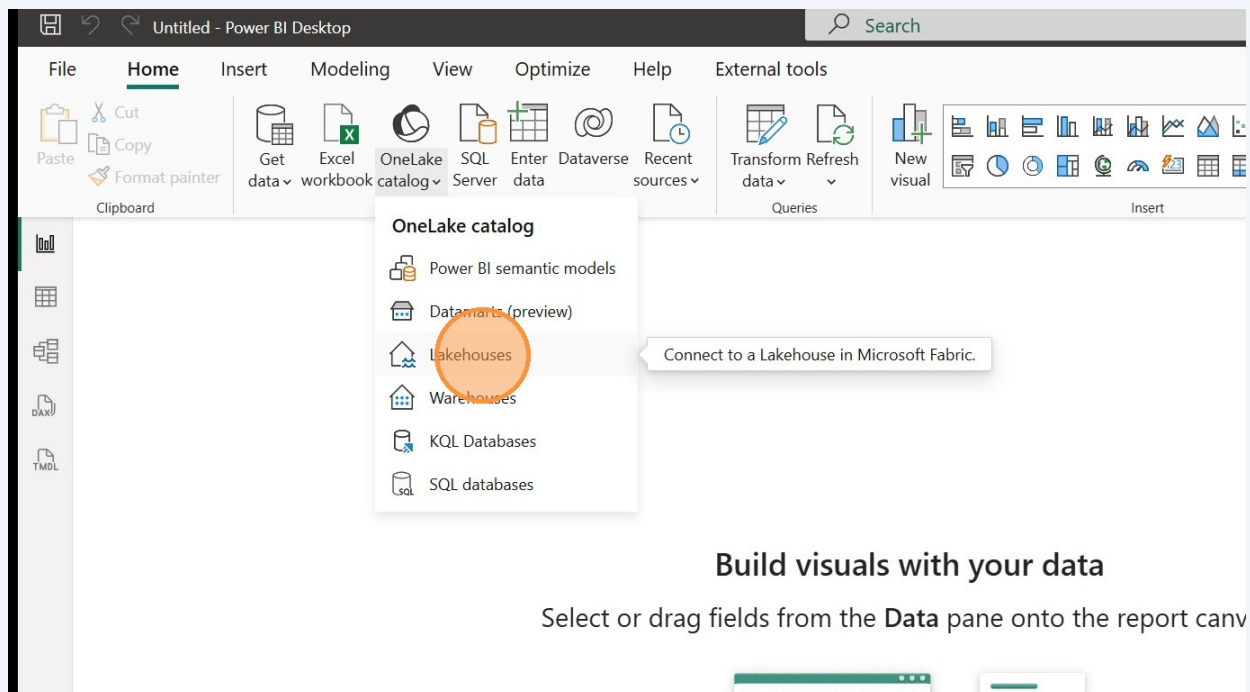


Tip! If you weren't able to get all the way through the Dataflows lab to get your four tables created, you can use this section to import those tables from the shared lakehouse instead.

9

There are two tables that our colleagues have created that we would like to use as well. We can simply import from their lakehouse instead of shortcutting.

Click the down arrow on the "OneLake catalog" button and select "Lakehouses."



10 Select the "sourcefiles_atl_fabcon" lakehouse.

OneLake catalog

Discover data from your org and beyond and use it to create reports

All My data Endorsed in your org External data Filter(1) ▾

Name	Owner	Refreshed	Location	Endorsement	Sensitivity
superfun_lakehouse	PBIF Test User 20	—	workshop tester	—	—
fabcon_sessions	Stephanie Bruno	—	lakshmanie	—	—
sourcefiles_atl_fabcon	Stephanie Bruno	—	PBIF shared	—	—
StagingLakehouseForDataflows_20...	PBIF Test User 20	—	workshop tester	—	—

[Learn more about connection options](#)

Connect Cancel

11 Click "Connect"

OneLake catalog

Discover data from your org and beyond and use it to create reports

All My data Endorsed in your org External data Filter(1) ▾

Name	Owner	Refreshed	Location	Endorsement	Sensitivity
superfun_lakehouse	PBIF Test User 20	—	workshop tester	—	—
fabcon_sessions	Stephanie Bruno	—	lakshmanie	—	—
sourcefiles_atl_fabcon	Stephanie Bruno	—	PBIF shared	—	—
StagingLakehouseForDataflows_20...	PBIF Test User 20	—	workshop tester	—	—

[Learn more about connection options](#)

Connect Cancel

12

OPTION 1: Use this if you were able to complete the Dataflow lab and have your four tables in your lakehouse.

Check the **two** checkboxes next to both the gold.conferences and gold.date table and click "Load."

☐ bronze.session_speakers_26_atlanta
☐ bronze.session_tracks_25_vienna
☐ bronze.session_tracks_26_atlanta
☐ bronze.sessions_25_vienna
☐ bronze.sessions_26_atlanta
☐ bronze.speakers_25_vienna
☐ bronze.speakers_26_atlanta
☒ gold.conferences
☒ gold.date

1/5/2026	2026	1	January	Jan	J
1/6/2026	2026	1	January	Jan	J
1/7/2026	2026	1	January	Jan	J
1/8/2026	2026	1	January	Jan	J
1/9/2026	2026	1	January	Jan	J
1/10/2026	2026	1	January	Jan	J
1/11/2026	2026	1	January	Jan	J
1/12/2026	2026	1	January	Jan	J
1/13/2026	2026	1	January	Jan	J
1/14/2026	2026	1	January	Jan	J
1/15/2026	2026	1	January	Jan	J
1/16/2026	2026	1	January	Jan	J
1/17/2026	2026	1	January	Jan	J
1/18/2026	2026	1	January	Jan	J
1/19/2026	2026	1	January	Jan	J
1/20/2026	2026	1	January	Jan	J
1/21/2026	2026	1	January	Jan	J
1/22/2026	2026	1	January	Jan	J
1/23/2026	2026	1	January	Jan	J

Select Related Tables

Load

Transform Data

Cancel

13

OPTION 2: Use this if you were **not** able to complete the Dataflow lab and do not have your four tables in your lakehouse. You can use these instead.

Check the checkboxes next to the **six** "gold." tables and click "Load."

☐ bronze.session_speakers_26_atlanta
☐ bronze.session_tracks_25_vienna
☐ bronze.session_tracks_26_atlanta
☐ bronze.sessions_25_vienna
☐ bronze.sessions_26_atlanta
☐ bronze.speakers_25_vienna
☐ bronze.speakers_26_atlanta
☒ gold.conferences
☒ gold.date
☒ gold.session speakers
☒ gold.session tracks
☒ gold.sessions
☒ gold.speakers

•

Preview is evaluating...

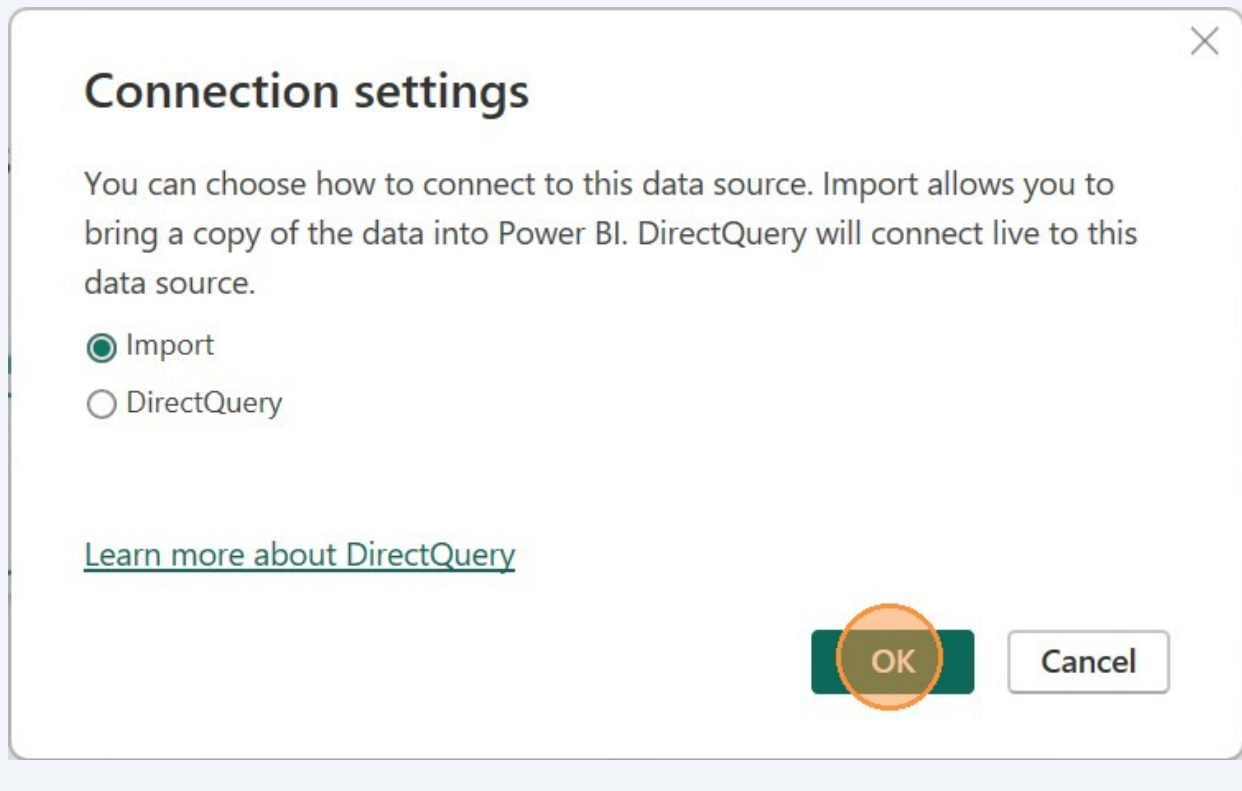
Select Related Tables

Load

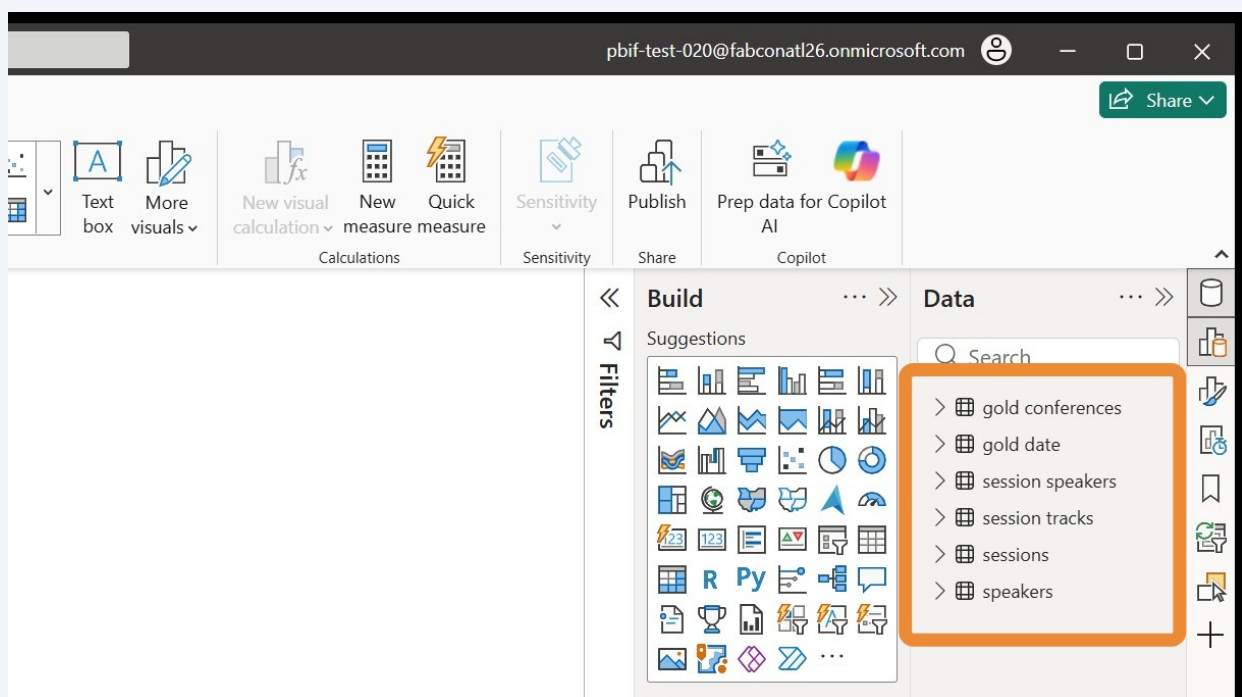
Transform Data

Cancel

- 14 Leave the default "Import" selected and click "OK."



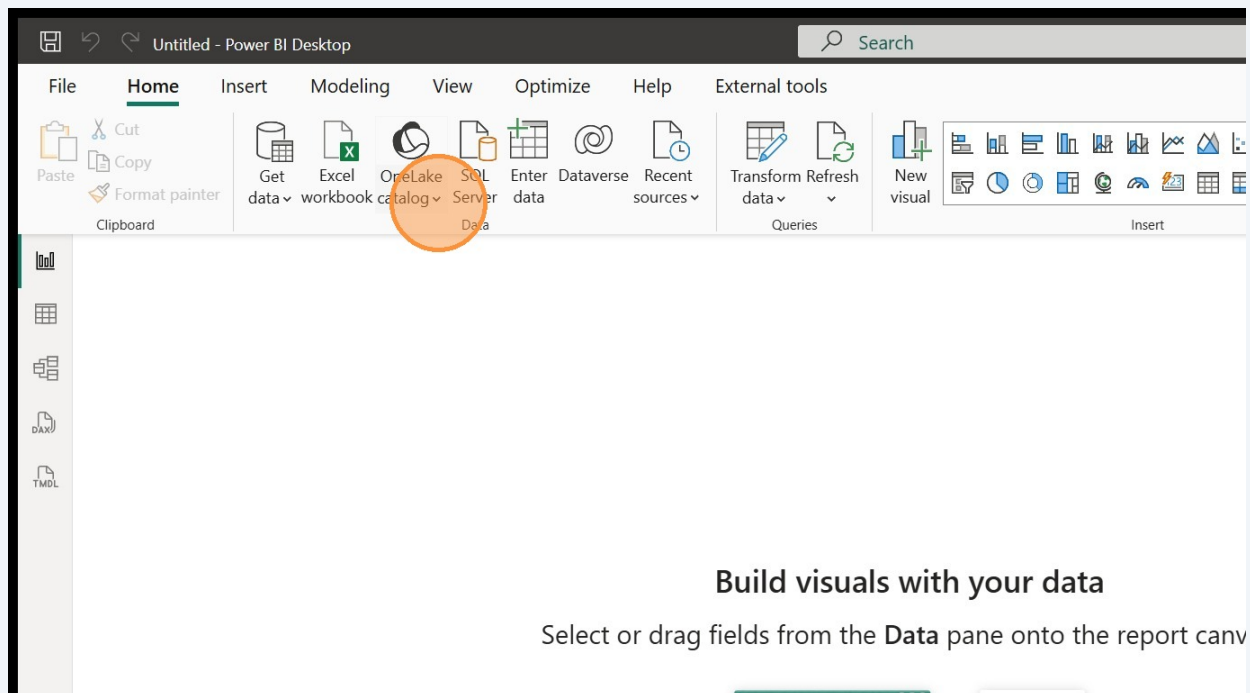
- 15 You should now have six tables loaded. They may or may not have a "gold " prefix. Either way is fine and we'll clean up the table names in a bit.



Connect to the SQL database - DirectQuery mode

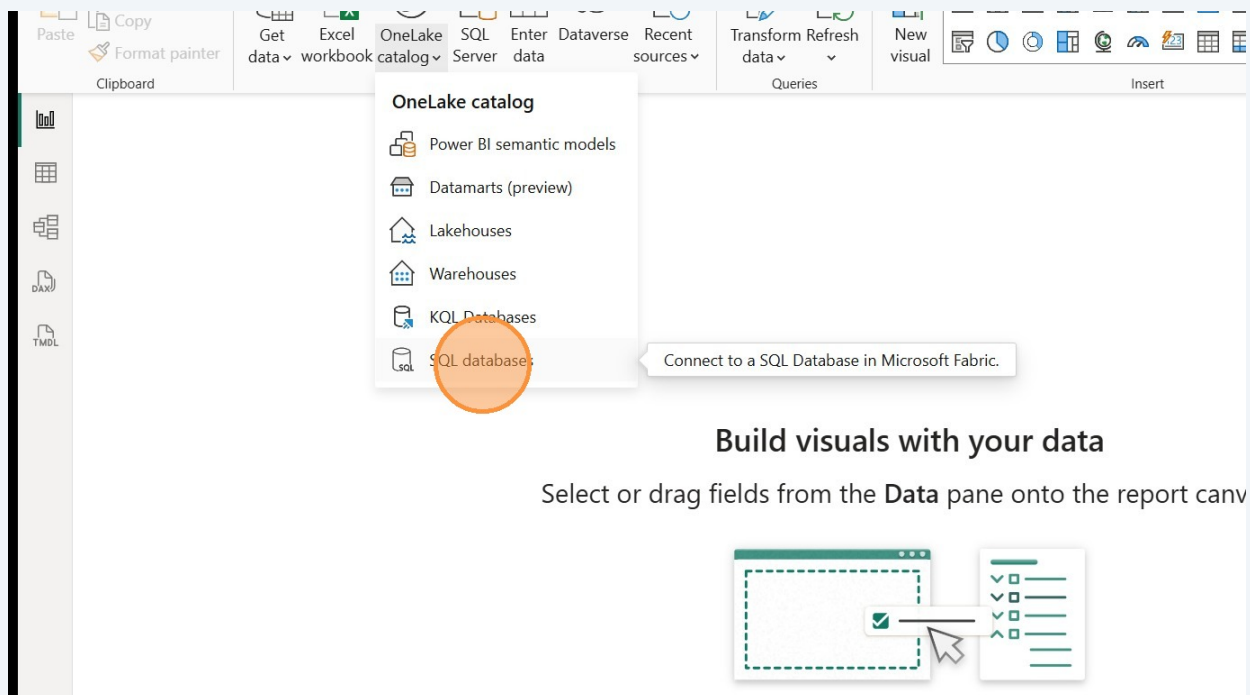
16

Now we'll get the attendees table from the SQL database. Click the **down arrow** next to "One Lake catalog" button in the Home tab (If you accidentally click the button instead, you'll just get a longer list of Fabric items to choose from, which is fine).



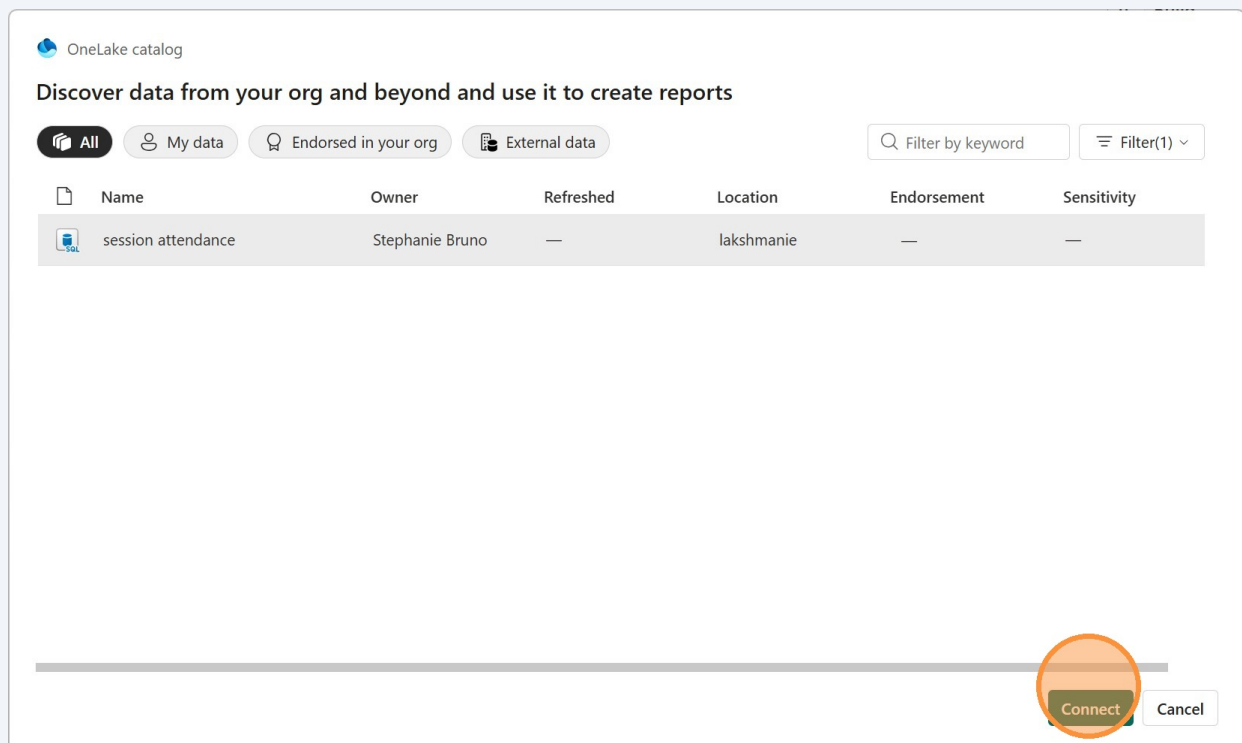
17

Choose "SQL databases." This will make it easier to find what you're looking for, especially if you have created a lot of items in Fabric.



18

Now navigate to your SQL database and click it to select it. Click the "Connect" button.



19 Check the box next to the "attendees" table and click the "Load" button.

The screenshot shows the Power BI Desktop interface. On the left, the 'Navigator' pane is open, displaying a list of tables. The 'attendees' table is selected, indicated by a green checkmark in a box next to its name. Below the table list, there is a 'Select Related Tables' button. On the right, the 'attendees' table is previewed, showing columns 'session_id', 'Attendee', and 'Status'. A message states 'This table is empty.' At the bottom right, the 'Load' button is highlighted with an orange circle. Other buttons visible are 'Transform Data' and 'Cancel'.

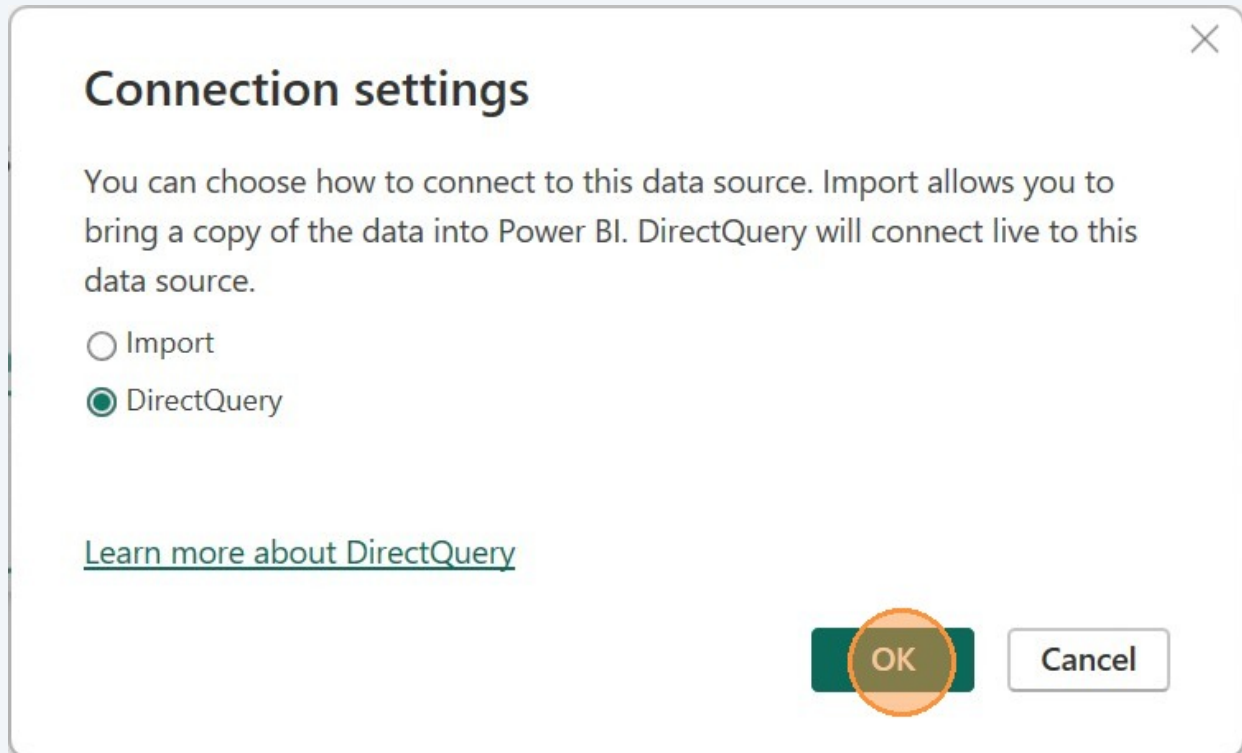


Alert! Make sure to choose DirectQuery for the next step.

This lets us see the results of our write-back functionality right away in the report, without requiring a refresh of the model. If you accidentally choose import, delete the table and reconnect in DirectQuery mode.

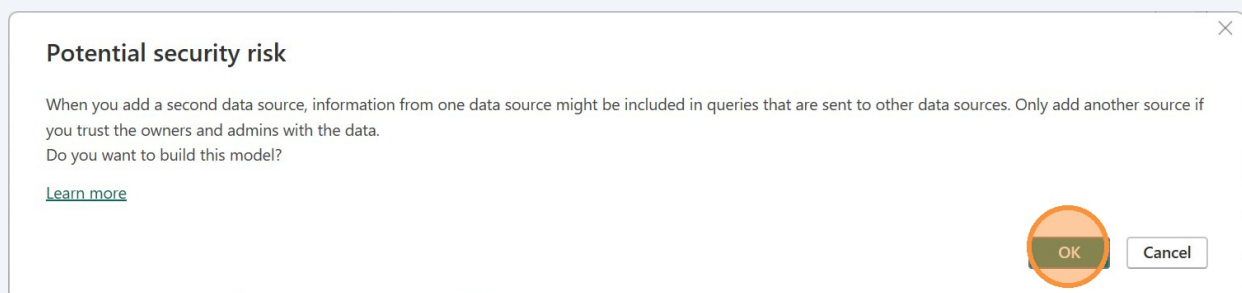
20

For this table, we want the mode to be "DirectQuery" so we can do our write-back and have the updates automatically show in the report. Click "DirectQuery" and then the "OK" button.



21

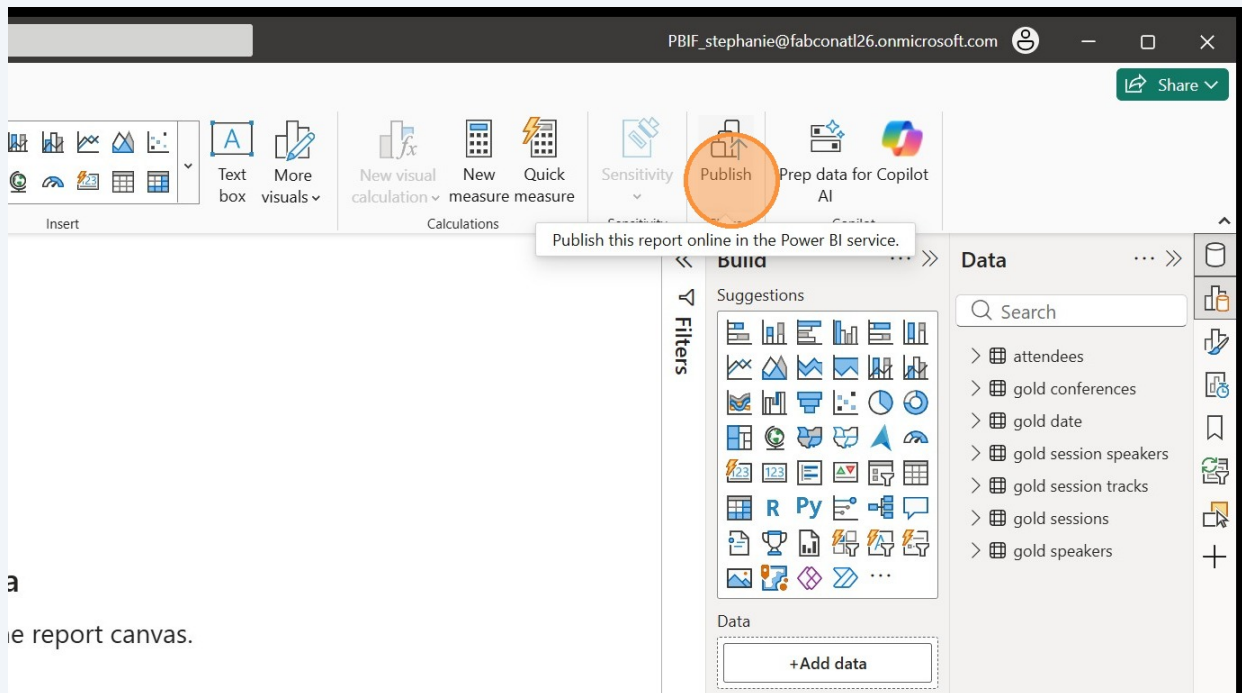
This message is letting you know that you have two different data sources that could potentially share data. Click "OK"



Publish to Fabric

22

Now we have seven tables in a new model. We'll publish it to the Power BI service before making any more changes. Click the "Publish" button.



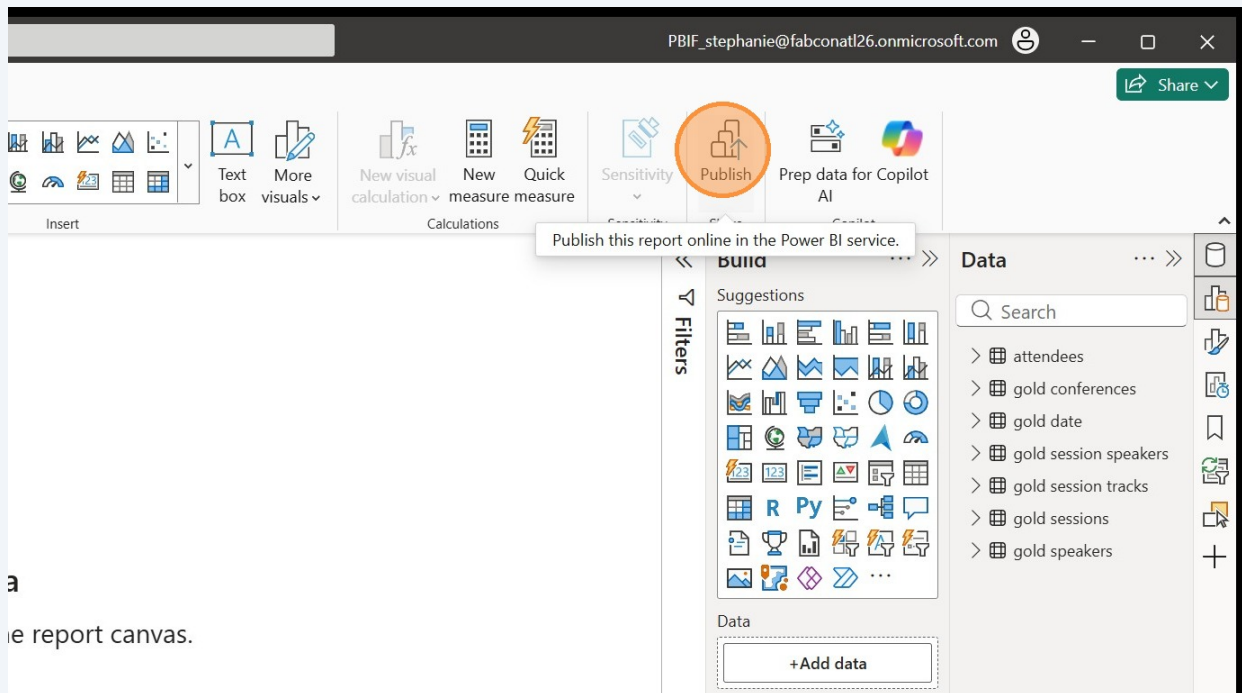
23

When you get a message asking you to Save the file, click the "Save" button and select a location.



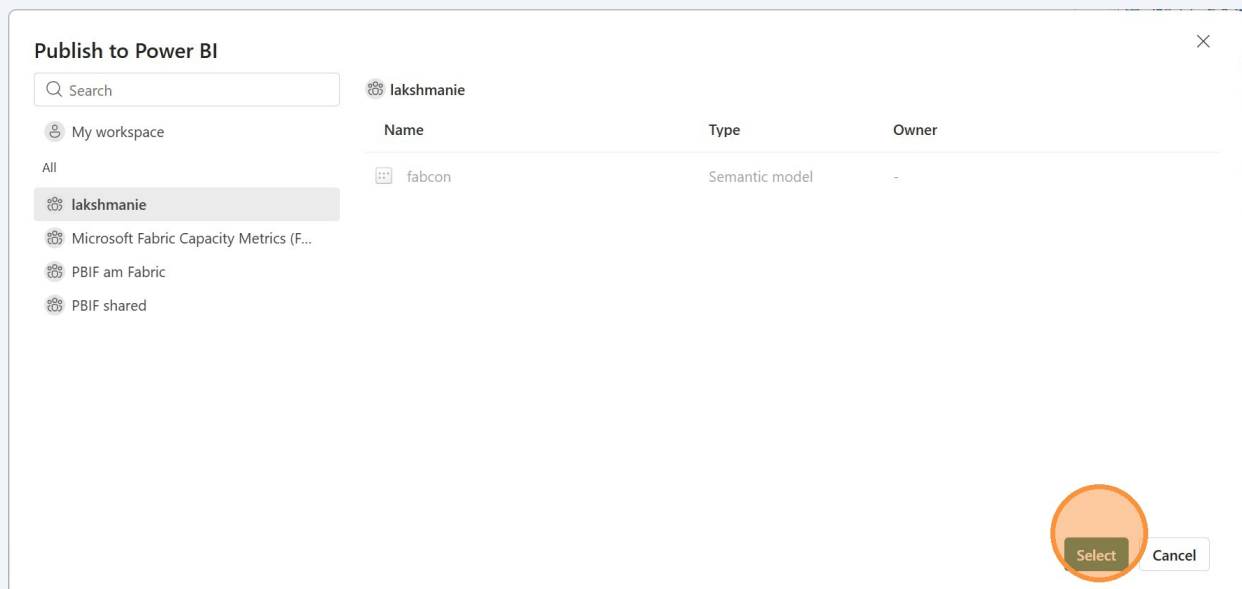
24

If the Publish window doesn't automatically pop up after you save the file, click the Publish button again.



25

Navigate to your workspace and click it, then click the "Select" button. This is where the model will be published.



26 Click "Got it"

Publishing to Power BI



✓ Success!

[Open 'fabcon sessions with attendees.pbix' in Power BI](#)



Did you know?

You can create a portrait view of your report, tailored for mobile phones. On the **View** tab, select **Mobile Layout**. [Learn more](#)

Got it

27 Great! Now we have a starter semantic model published to the workspace. We'll continue to edit it in the workspace. We're finished with Power BI desktop now so you can close it if you like.